

Executive Summary

Swirltubs is confronted with ongoing inefficiencies by its 1,000 warranty repair technicians that operate independently to manage van inventory, which normally leads to slow services whenever critical parts are out of stock. With an annual inventory carrying cost of 25% and \$50 hourly technician labor, the company needs to effectively inventory more than 400 high-usage parts in a 500 cubic feet van capacity. A heuristic approach was used in modeling the uncertainty of demand and optimizing the most cost-efficient mix of parts for mixed technician demands. In 100 random instances, a total average of 213 parts were chosen per van for a predicted first-visit rate of just under 70% at an average cost of \$14,279.90. This strategy offers a cost-effective databased inventory policy that responds to demand volatility while balancing cost and high service performance.

Business Background

Swirltubs is a big appliance company with a national service warranty network of 1,000 field technicians. There are about 500 cubic feet per technician of inventory in vans. Decentralization has led to high inventory carrying costs (25% annually) and service delivery inefficiencies where 22% of the repair have return trips due to lack of parts. Although a data-driven, centralized stocking system seemed plausible, Swirltubs now need to factor in the additional demand uncertainty within and across time among technicians. To create an affordable, scalable inventory plan, Swirltubs will need to implement an adaptive stocking strategy that adjusts to uncertain, technician-dependent patterns of part usage.

Problem Background

Swirltubs technicians currently buy and maintain their own van stock, creating inconsistency and inefficiency fleet wide. If the parts are not on-hand when the service call is received, technicians must order and come back later resulting in tardy repairs, irate customers, and decreased productivity. With every return is added labor and travel expense, decreasing weekly repairs. While Jim Jenkins, manager of Aftermarket Logistics, did come up with a stocking formula based on part usage, cost, and size, it is based on constant demand across technicians and in the long run. Part failure is extremely stochastic. A performing stocking plan for one technician would not be so for another. To deal with this, a more dynamic, scenario-based method needs to be used to model how part selection is impacted by uncertainty in technician-specific demand.

Problem Statement

Swirltubs need to create a data-driven, centralized van inventory replenishment strategy within the capacity of each van's 500 ft³ that reduces total cost of service is 25% annual carrying cost of inventory and expensive technician re-visits while enhancing first-visit repair achievement. The solution needs to allow for technician-level variation in part requirements. In addition to merely optimizing a given stock list, the firm also needs to ascertain how well the decisions to stock fare under uncertain, technician-part-specific demand to keep the plan successful under a broad variety of realistic conditions.

Assumptions

1. Single-Part Repairs: One part per repair is enough. Assembly pieces are counted as a single unit of inventory.
2. Five-Day Replenishment: All non-inventory parts that are sent from the central warehouse materialize in five workdays.
3. Van Capacity Constraint: Every one of the technician vans has a 500 cubic feet inventory capacity for the whole dataset analysis; 50 ft³ was utilized in the preliminary test dataset analysis.
4. Labor Cost: Technician travel and waiting time are billed at \$50 per hour.
5. Inventory Holding Cost: There is a holding cost of 25% of the value of each part per year.

6. Representative Dataset: The 416-part dataset approximates about 80% of the typical technician-level repairs and is based on historical network-wide demand.
7. Uniform Demand Baseline: Since the initial model assumes uniform use of parts across technicians, Part 5 adds random variation to model uncertainty at the technician level.
8. No Geographic Restriction: Swirltubs has no geographical restriction on part stocking, and failure rates are assumed uniform over regions.
9. Work Schedule: Technicians are present 250 workdays a year; difference in work schedules is not captured in modeling.
10. Stochastic Part Failure: Part failures are stochastic, modeled based on past usage patterns and randomized binomial distributions.
11. Technician Compliance: Technicians can be taken to consent and subscribe to a standardized stocking plan if it clearly enhances service performance and operating efficiency.
12. Assumption of Data Accuracy: Historical usage data used to simulate demand is presumed to be accurate and reflective of future trends of breakdown. Unforeseen product line or customer usage pattern changes are not accounted for.

Scope

This study focuses on assessing the degree to which the recommended stocking policies are still under the situation of technician-level variation and ambiguous part demand. Contrary to the preceding stages that had used average assumption of use, this stage applies stochastic simulation to replicate 100 various scenarios of demand using binomial distributions based on historical utilization. In each simulation, part requirements were randomly set and subsequent revisit costs recalculated so that reselection of parts on a heuristic score basis could occur. The optimization phase checks if prior stock decisions are stable under varied conditions and what parts are resiliently chosen vs. demand dependent. The scale of this phase remains concentrated on the 416part set and a 500 ft³ van volume but encompasses an evaluation framework based on scenarios that brings more confidence to the plan's scalability, flexibility, and reliability for long-term deployment in the real-world.

Data

This study utilizes the sample 416-piece data set for about 80% of Swirltubs' yearly repair business. Unit cost in dollars, physical size in cubic feet, and historical usage frequency were on each piece record. Obsolete or infrequently used pieces were excluded. Historical use was the basis during this stage to generate randomized technician-level demand for 100 scenarios. Computer-based demand enabled recalculation of revisit cost and optimization of van stock regarding 500 ft³ capacity constraint.

Method/Approach

This stage of the project establishes the stability of the variable and uncertain part demand van stocking plan. Based on previous models, we generate 100 technician-level demand scenarios and re-opt parts dynamically for each scenario. The research employs four inventory models for comparison:

1. Optimization Model

A binary integer programming formulation minimizing total cost as the sum of inventory holding cost (25% of part cost) and return visit labor cost (\$50/hour) subject to 500 ft³ van space availability. It was first tested on the 30-part subset and then scaled up to the full 416-part sample to provide an upper-bound estimate of bestcase performance.

2. Jenkins Threshold Model

This cutoff-based method chooses pieces with clear-cut thresholds: usage > 1, cost < \$70, and size < 8 ft³. It has a simple, easy-to-implement policy for stocking and was a good choice to begin with heuristic-based decision-making.

3. Heuristic Model Variants

Other threshold-tuned heuristics (e.g., usage > 1.5, cost < \$150, size < 5.5 ft³) and ranking methods were also attempted. These models sacrifice simplicity to deliver better service performance via selection of parts according to predefined business rules.

4. Optimization Heuristic

This model recycles the same cost-efficient formula of previous phases:

$$(Revisit\ Cost - Inventory\ Cost) \div Part\ Size$$

But during this stage, simulation usage of parts is randomized within every simulation of historical average demand with a binomial distribution. Return cost for every simulation is re-computed as per simulated usage, and the parts are scored again. The parts with the highest scores are picked up until capacity of 500 ft³. This dynamic selection of parts enables variability of demand technician-to-technician to be supported by the model.

Simulation Design

Random part usage was constructed over more than 100 runs based on binomial distributions scaled to the mean annual usage of each component. Revisit cost was then re-calculated using:

$$Revisit\ Cost = 25 \times (1 - Simulated\ Use \div 5)$$

This allowed revisit penalty to be dynamic per run depending on random component failures.

Selection Logic and Evaluation Metrics For every run:

- Parts were ranked on cost-effectiveness score
- Selected randomly until total space used up to 500 ft³
- A binary choice matrix was constructed (1 = selected, 0 = not) Every simulation was evaluated with:
 - Inventory Cost: Total of holding cost for selected parts
 - Revisit Cost: Total of revisit penalty of unselected but needed parts
 - Total Cost: Inventory and revisit combined
 - First Visit Success Rate: % of total simulated demand met by selected parts

This approach allows us to test the consistency and quality of the heuristic approach over real uncertainty and identify which parts are consistently selected against being sensitive to technician-level demand variation.

(Anticipated) Results

The table below summarizes the performance of the optimization and heuristic approaches based on actual results from the Excel models:

Model	Parts Selected	Total Volume Used	Estimated Cost (\$)	Notes	First Visit Success Rate
Optimization Model	7	49.98 cubic feet	\$1,556.14	Chose only a few parts based on lowest cost; didn't hold up in practice	21%

Jenkins Model Test Data	10	47.76 cubic feet	\$875.57	Used fixed rules (usage > 1, cost < \$70, size < 8 ft ³); not tested in simulation	40%
Heuristic Model Test Data	16	40.81 cubic feet	\$686.26	Balanced usage, size, and cost; gave the best performance	67%
Heuristic Model Raw Data	213	494.60 cubic feet	31,749.47	Applied to full dataset; realistic performance based on actual demand	47%
Jenkins Model Raw Data	161	480.49 cubic feet	\$29,981.72	Applied full rule set to complete dataset; performed moderately well under real conditions	43%

- **Optimization Model:** Had the lowest theoretical total cost but a low 21% first-visit rate under simulated demand. The model strongly rewarded cost savings and did not retain enough of the most in-demand parts, with an extremely high revisit rate and less service reliability.
- **Jenkins Threshold Model:** Was a low-level benchmark with simple usage, cost, and size thresholds. Although basic and easy to implement, not very flexible, only offered mediocre service coverage, and thus less suitable under risky demand conditions.
- **Heuristic Model:** Provided a good mix of service and price. With rules aimed at the most used, smallest, and cheapest parts, it achieved a 67% first visit hit rate and eliminated a lot of return visitation. It needed no optimization software either, so roll-out in the field was an option.
- **Composite Heuristic (Cost-Efficiency Oriented):** This heuristic consistently ranked higher than others based on coverage, cost-efficiency, and usability. It never required sophisticated tools and chose components through a dynamic scoring approach, and it performed the best overall on the test set.
- **Role of Simulation:** Conclusions reveal theoretical models fall short. Randomized part demand testing vs. found service reliability is enhanced by accounting for variation. This emphasizes that optimization and simulation should be core components of stocking solutions.
- **Equivalence in Testing:** Models were tested on a 50 ft³ constraint and sampled 30-part case to facilitate equal comparison.
- **Final Heuristic Stock Plan:** Executed on the full dataset, the final model chose 213 items, consumed 494.6 cubic feet and had a 47% first-visit success rate. Inventory cost was \$3,566.91, revisit cost was \$10,712.99, and total cost per van was \$14,279.90, showing a good baseline with room for additional tuning.

Total Cost Distribution:

Model	Avg Holding Cost	Avg Revisit Cost	Total Cost Range
Optimization Model	\$500	\$1,000 – \$1,300	\$1,500–\$1,800
Heuristic Model	\$3,566.91	\$10,712.99	\$14,000 – \$14,500

- The original optimization model was cheap for inventory but captured wildly erratic revisit costs occasionally over \$1,300, because it had unavailable high-demand parts under typical technician usage.
- The reoptimized heuristic model, pilot-tested on 100 randomly simulated demand scenarios, was less erratic. Though it was more costly in inventory expense, it offered more part coverage and greatly fewer revisit occurrences.
- It consistently chose 213 parts, used 494.6 cubic feet of van space, and had a consistent total cost of \$14,279.90 in \$14,000–\$14,500 range for each simulation.
- The model's reliability in uncertainty demonstrates its strength and makes it a more realistic choice for large-scale, variable technician environments.

Recommendations:

- Use the optimization model tactically as a cost-effectiveness benchmark, but not operationally because it is unresponsive under changing demand.
- Use the Jenkins threshold model as a fallback for fast, rule-based inventory decisions in situations where time or resources are constrained.
- Use the reoptimized heuristic model (213 items, 494.6 ft³, \$14,279.90) as the main stock plan, because it performed consistently well in 100 simulated demand environments.
- Base the first-visit rate at 47% and continuously adjust data-driven coverage enhancement and growing part sets over time.
- Test the final parts list in 25–50 technicians to measure in-the-field service effectiveness and gain feedback for subsequent releases.
- Phase in the stocking plan beginning in those markets that have mature inventory support systems to make adoption smoother.
- Offer technician training to clarify the advantages of the centralized plan and raise part usage awareness.
- Offer technicians a portion of van space (e.g., 5–10%) to determine. This provides flexibility and encourages technician participation without sacrificing central control.
- Revise the inventory strategy every 3–6 months by utilizing technician use, service patterns, and other simulations to keep the stocking plan aligned with actual demand.

Potential Outcomes:

- 10–20% savings on inventory cost across vans by more focused, low-cost stocking strategy.
- Higher first-visit repair rates, reducing delays and improving customer satisfaction.
- Fewer return trips and reduced downtime for technicians, particularly versus the 21% potential using cost-based optimization models.
- More consistent inventory across the fleet, reducing restocking, training, and control complexity.

- Fewer service delays due to unavailable high-demand parts, although occasional re-visits due to seldom available parts still happen.
- Enhanced service performance stabilized at the regional level, with the heuristic model more tolerant of variability in technician demand.

Risks

- Part Preference Match-Up: Although technicians would most prefer individually customized stocking, this has been offset by maintaining some van space to accommodate parts selected by the technician.
- Change of Repair Patterns: Appliance defect patterns change over time, so this will limit the effectiveness or relevance of a fixed list of parts.
- Demand Unpredictability: Projected demand will probably fall short of capturing escalating service needs, leading to periodic shortages of unexpected parts.
- Van Space Constraints: Physical storage capacity could be otherwise constrained by inefficiencies in shelving, tooling, or packaging not represented.
- Model Inflexibility: Rule-based stocking approaches might not be able to accommodate atypical technician areas or one-service demands with ease.
- Stale Plans: The model for stocking can become outdated over time as service trends change if it is not periodically refreshed.
- Service Risk via Cost Emphasis: The 21% first-visit rate of the optimization model also points to the risk of cost emphasis at the cost of increased return visits and labor expense.
- Service Quality Trade-offs: Cost-reduced models sacrifice parts availability, resulting in lost repairs and customer dissatisfaction.

Next Steps

- Pilot Program: Pilot run the reoptimized heuristic model for 60 days in 25–50 technicians. Monitor first-visit success, revisit cost, inventory usage, and comments from technicians to verify operation under varying demand in actual conditions.
- Offer Technician Flexibility: Leave 5–10% of van capacity for parts specified by the technicians to ensure maximum take-up, confidence, and provision for local or situation-based requirements.
- Build Technician Training: Develop rapid-start guides and in-field training to describe how the stocking model operates, why it enhances service consistency, and how it enables technician autonomy.
- Institute a Regular Update Schedule: Plan quarterly updates to the stocking model based on new demand data, repair patterns, and technician usage to keep it aligned with changing field conditions.
- Integrate into Inventory Systems: Start integrating reoptimized stock list with ordering and restocking systems to automate van inventories management and eliminate manual mistakes.
- Stress-test against extreme scenarios: Continue model testing under edge-case or nonhistorical demand conditions to ensure robustness and readiness against emergent breakdown patterns.
- Track Core KPIs: Deploy performance dashboards to track first-visit success rate, average total cost, van space usage, and revisit frequency as improvement components on a continuous basis.